

- N.B.**
- 1) All questions are compulsory.
 - 2) Figures to the right indicate marks.
 - 3) Illustrations, in-depth answers and diagrams will be appreciated.
 - 4) Mixing of sub-questions is not allowed.

Q.1 Attempt the following (Any FOUR)(Each of 5Marks)**20 Marks**

- a. What is Algorithm? Explain goals of analysis of algorithm.
- b. What are different types of Data Structures?
- c. What are advantages and disadvantages of Arrays?
- d. Write short note on Big-O notation.
- e. How to represent Polynomials using Linked List? Consider following polynomial equations and represent in Linked List.

$$x^2 - 3x + 2 = 0$$

$$3x^3 + 2x + 2 = 0$$

$$4x^4 - 3x^2 + 3x + 1 = 0$$

- f. Convert following expressions from Infix into Postfix notation step by step.

- i. $A+B-C*A+C/D$ ii. $2+4/5*(5-3)+5-4$

Q.2 Attempt the following (Any FOUR) (Each of 5Marks)**20 Marks**

- a. Differentiate between Recursion and Iteration.
- b. Consider following members and sort using Bubble Sort.
5, 25, 15, 10, 22, 17, 20, 35, 30, 40
- c. What is Searching? Write an algorithm for Binary Search.
- d. What is String? Consider following string $T=ababaabababacaababa$ and Pattern $P=abab$ apply Pattern Matching technique and find the positions where pattern get recognized.
- e. Write an algorithm for finding sum of all integers from 1 to n using iteration.
- f. What is Selection? Write algorithm for Kth Smallest elements in sorted order.

Q.3 Attempt the following (Any FOUR) (Each of 5Marks)**20 Marks**

- a. Explain various Algorithm design techniques of classification by Implementation Method.
- b. What is Greedy Strategy? Explain different elements of Greedy Algorithm.
- c. What are advantages and disadvantages of Divide-n-Conquer method?
- d. Consider following matrix A and B and find multiplication using Strassen's Matrix multiplication method.

$$\begin{bmatrix} 1 & 7 \\ 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} 3 & 3 \\ 5 & 2 \end{bmatrix}$$

A B

- e. Explain Dynamic Programming. What are different approaches of Dynamic Programming?
- f. What is Backtracking? Solve n-Queen problem where $n=4$ using backtracking.

Q.4 Attempt the following (Any FIVE)(Each of 3Marks)

15 Marks

- a. Write an algorithm to find largest among three numbers.
- b. Write short note on Time complexity.
- c. Explain comparative analysis between Bubble, Insertion, Selection Sort.
- d. What is Transform and Conquer?
- e. List down Applications of Greedy Techniques.
- f. What are techniques to solve Dynamic Programming?

munotes.in